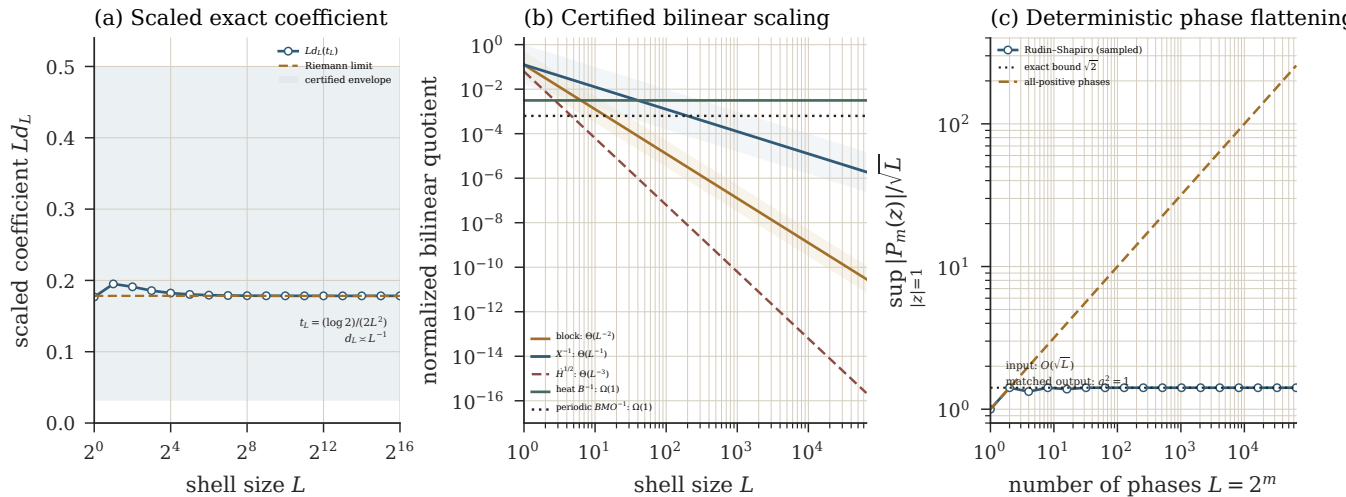


Exact Duhamel denominator: shell gain versus critical saturation



Exact output: $d_L(t) = e^{-t} \sum_{N=L}^{2L-1} (1 - e^{-2N^2 t}) / (2N^2)$. At $t_L = (\log 2)/(2L^2)$, $1/(32L) \leq d_L(t_L) \leq 1/(2L)$ for every L .

Formal scope: 8,390,656 packet modes, Rudin-Shapiro length 4,194,304, 24/24 checks; exact integer and $Q(\sqrt{2})$ audit. Curves and sampled maxima are presentation data.

Conclusion: the time integral gives a genuine heat denominator, but critical negative regularity and phase flattening can consume its shell gain. This is not norm inflation or a Navier-Stokes regularity proof.